

Exit Ticket Questions for Week 4

1. Why does a symmetric FIR filter with N taps give constant group delay?
2. Why does the bilinear transform preserve stability?

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}} \Rightarrow z = \frac{1 + \frac{sT}{2}}{1 - \frac{sT}{2}}$$

Good answers:

1. Symmetric means $h[n] = h[N-1-n]$. Shift left by $(N-1)/2$ to obtain $a[n] = h[n + (N-1)/2] = h[(N-1)/2 - n] = a[-n]$. Because of the symmetry of $a[n]$, its DTFT $A(e^{j\omega})$ must be real-valued and thus $H(e^{j\omega}) = A(e^{j\omega}) * e^{-j\omega(N-1)/2}$. The group delay is the negative derivative of the phase $-\omega(N-1)/2$ wrt ω and is thus the constant $(N-1)/2$.

Shifting the center of symmetric FIR filter to $n=0$ creates an impulse response with even symmetry. Its DTFT must be real-valued and thus the only phase comes from shifting the impulse response. This phase is linear in ω and its derivative is thus a constant, resulting in a constant group delay.

Look for shifting center to zero, creating symmetric impulse response whose DTFT must be real-valued.

2. If $\text{Re}\{s\} < 0$, then the inverse bilinear map has a numerator with smaller magnitude than the denominator and thus $|z| < 1$ and the left-half s -plane maps into the unit circle in the z -plane.

Look for $\text{Re}\{s\} < 0$ as the criterion for $|z| < 1$ and the use of the inverse bilinear transform.

A good exit ticket answer should be short, correct, mathematically sound, to the point and demonstrate a well-reasoned understanding of the subject that the question is asking about. Exit ticket answers will be graded on a scale of “low” (1 point), “medium” (2 points), or “high” (3 points).

Use LaTeX style for math expressions, e.g., $\delta[n]$ for a discrete-time impulse, or use the math equations in the rich content editor in Canvas.